

## Algorithm for Fixed-Point Iteration

- Input: Initial guess  $x_0$ , maximum iterations “steps”, tolerance  $\epsilon$ .
- Initialize:
  - Set  $x_r = x_0$  (current approximation).
  - Set iteration count  $n=1$ .
  - Set initial error  $er=1$ .
- Repeat Until Convergence or Maximum Iterations:
  - Store the previous approximation  $x_{r\_old}=x_r$ .
  - Compute the next approximation using  $x_r=g(x_r)$ .
  - Compute relative error:  $er=\frac{x_r-x_{r\_old}}{x_r}$
  - If  $er<\epsilon$ , stop.
  - Increment iteration count  $n$ .
- Output: Approximate root  $x_r$ .

- **Example-2**  $f(x)=2x^3-2x-5$   
Find a root of an equation  $f(x)=2x^3-2x-5$  using Fixed Point Iteration method

**Solution:**

Method-1

Let  $f(x)=2x^3-2x-5$

Here  $2x^3-2x-5=0$

$$\therefore 2x^3=2x+5$$

$$\therefore x^3=2x+5$$

$$\therefore x = \sqrt[3]{2x+5}$$

$$\therefore \phi(x) = 3\sqrt{2x+52}$$

Here

|             |    |    |   |
|-------------|----|----|---|
| <b>x</b>    | 0  | 1  | 2 |
| <b>f(x)</b> | -5 | -5 | 7 |

Here  $f(1) = -5 < 0$  and  $f(2) = 7 > 0$

$\therefore$  Root lies between 1 and 2

$$x_0 = 1 + \frac{2}{2} = 1.5$$

$$x_1 = \phi(x_0) = \phi(1.5) = 1.5874$$

$$x_2 = \phi(x_1) = \phi(1.5874) = 1.59888$$

$$x_3 = \phi(x_2) = \phi(1.59888) = 1.60037$$

$$x_4 = \phi(x_3) = \phi(1.60037) = 1.60057$$

Approximate root of the equation  $2x^3 - 2x - 5$  using Iteration method is 1.60057 (After 4 iterations)

| <b>n</b> | <b>x<sub>0</sub></b> | <b><math>x_1 = \phi(x_0)</math></b> | <b>Update</b> | <b>Difference<br/><math> x_1 - x_0 </math></b> |
|----------|----------------------|-------------------------------------|---------------|------------------------------------------------|
| 2        | 1.5                  | 1.5874                              | $x_0 = x_1$   | 0.0874                                         |
| 3        | 1.5874               | 1.59888                             | $x_0 = x_1$   | 0.01148                                        |
| 4        | 1.59888              | 1.60037                             | $x_0 = x_1$   | 0.0015                                         |
| 5        | 1.60037              | 1.60057                             | $x_0 = x_1$   | 0.00019                                        |